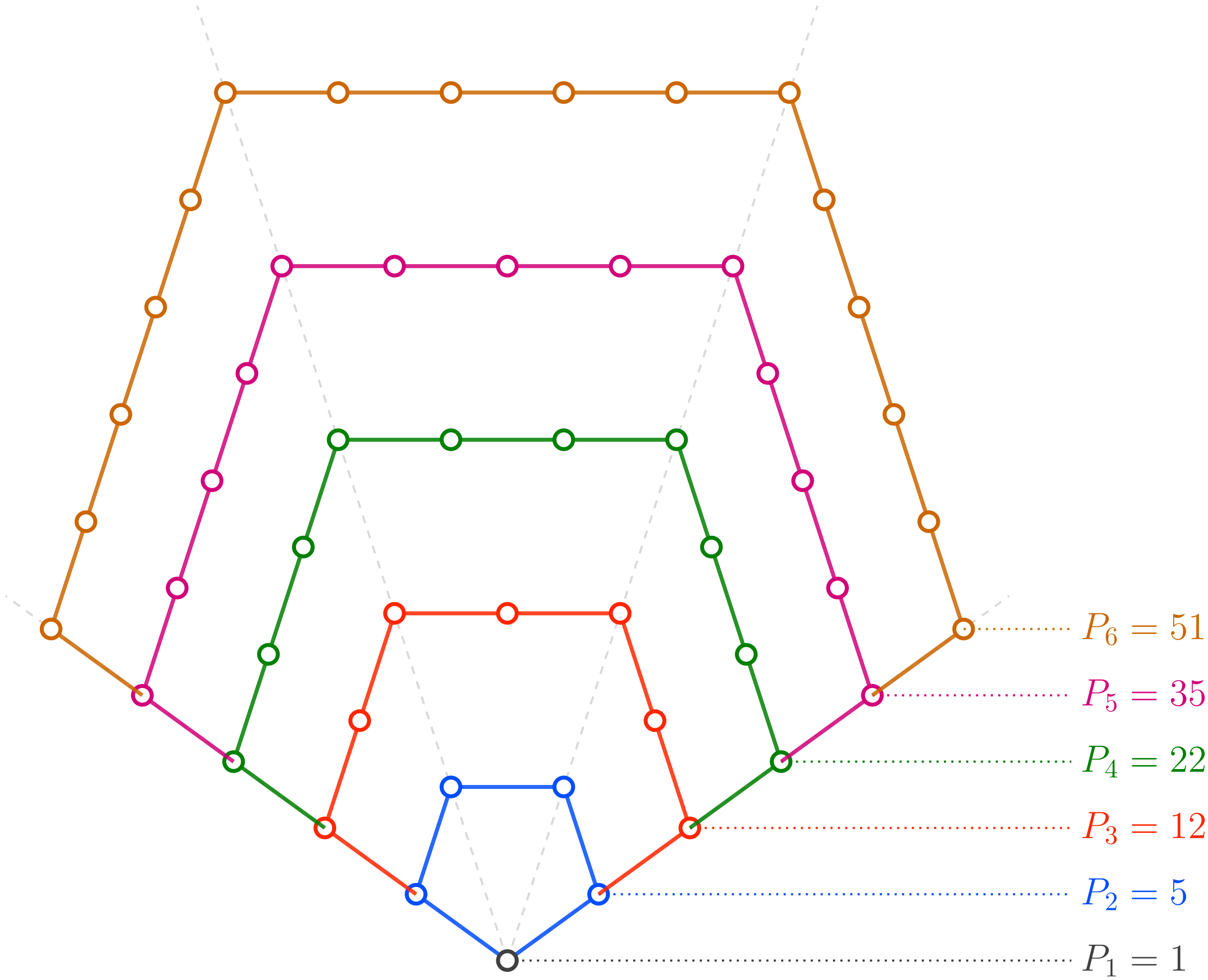

Euler's Pentagonal Number Theorem

$$\prod_{n=1}^{\infty}(1 - x^n) = \sum_{k=-\infty}^{\infty}(-1)^k x^{g_k}$$

$$= 1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + x^{22} + x^{26} - \dots$$

where $g_k = \frac{k(3k-1)}{2}$ are the **generalized pentagonal numbers**:

$g_k = 1, 2, 5, 7, 12, 15, 22, 26, 35, \dots$ for $k = 1, -1, 2, -2, 3, -3, 4, -4, 5, \dots$